

Optical Coatings

Optical coatings typically consist of thin films made up of single or multiple layers properly designed and fabricated, these coatings can dramatically modify the reflectance component. The properties can be controlled from the deep UV to the IR with narrow bandwidths and can be polarization sensitive. Optical coatings can be applied directly to the surface of a component in the case of an optical mirror or beamsplitter. For other components, such as lenses, they can improve overall transmission properties by reducing surface reflectivity. When optical coatings are used for the express purpose of controlling the spectral transmission of light, the component is called an optical filter.

Optical Coating Process

The individual layers that make up optical coatings are typically a few tens of nanometers thick. While a single optical coating can be comprised of several hundred layers. Consequently, the process requires a high degree of precision. Generally, the process begins with surface flatness and free of surface damage. It continues with surface cleaning and preparation and is followed by deposition according to design. The deposition technologies include thermal evaporation, electron-beam evaporation, sputter deposition. The most appropriate coating technology for the intended product depends on the material requirements, physical characteristics, application requirements, and economic considerations. Comprehensive performance testing using sophisticated metrology tools.

Types of Coatings

Metallic coatings used on optical mirrors typically consist of a single layer approx. 100 nm thick. The high reflectivity properties of the metal due to the complex index of refraction and absorption. For specific wavelengths of interest, dielectric coatings (also called interference coatings) consist of alternating high refractive index ($n_H = 1.8 - 4.0$) and low refractive index ($n_L = 1.2 - 1.5$) layers (see Figure 1). The thickness of each layer is chosen such that the optical path length is $\lambda/4$. A variation of the formula given by the Reflectivity Equation (see Optical Mirrors) shows that the reflectivity of the dielectric coating which increases with a greater number of layers and narrower spectral bandwidth. Dielectric coatings are ubiquitous and their applications range from anti-reflection to high reflectors. Metal-dielectric hybrid films such as those in cube beamsplitters and absorptive mirrors are also found in certain optical filters.

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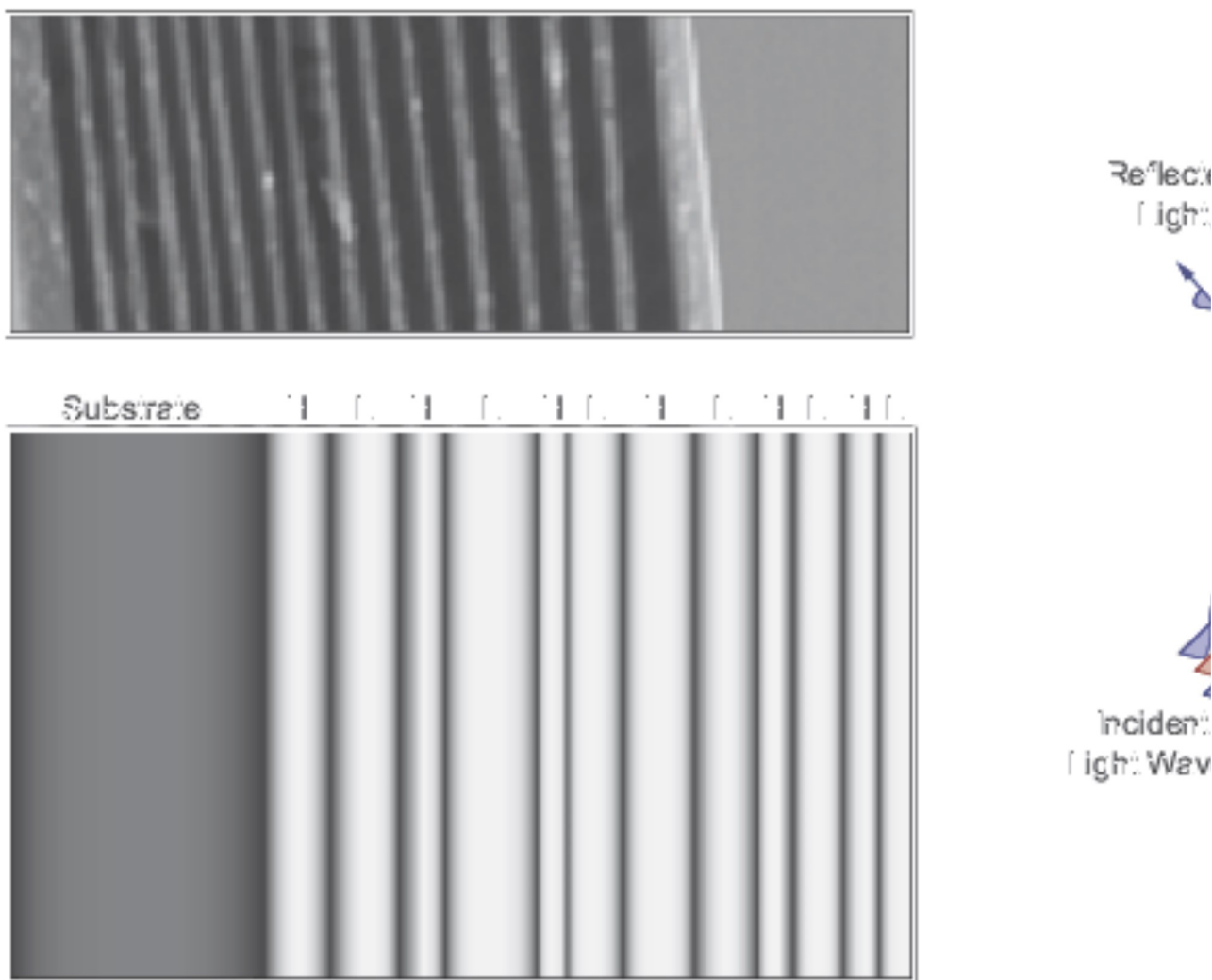


Figure 1. Scanning electron microscope image (top) and schematic (bottom) of an optical interference coating showing a filter consisting of an interference coating (right).

Dielectric optical coatings are used in a myriad of ways. In addition to highly reflective coatings, they are incorporated in broadband beamsplitters and dichroic filters (color-selective mirrors). When light is incident at an angle to a surface, i.e., not normal incidence, the coating is sensitive to polarization. This allows dielectric coatings to be polarization selective and such coatings can be used to enhance or reduce reflectivity. To enhance the reflectivity, dielectric optical coatings can also be used to reduce reflection coatings. These coatings can be applied to any optical component, e.g., a lens, to improve its transmission efficiency. Based on the reflectivity equation, the reflectivity of a single layer coating can be reduced considerably with a broadband anti-reflection coating. This superior performance, however, is achieved at the expense of increased thickness and cost.

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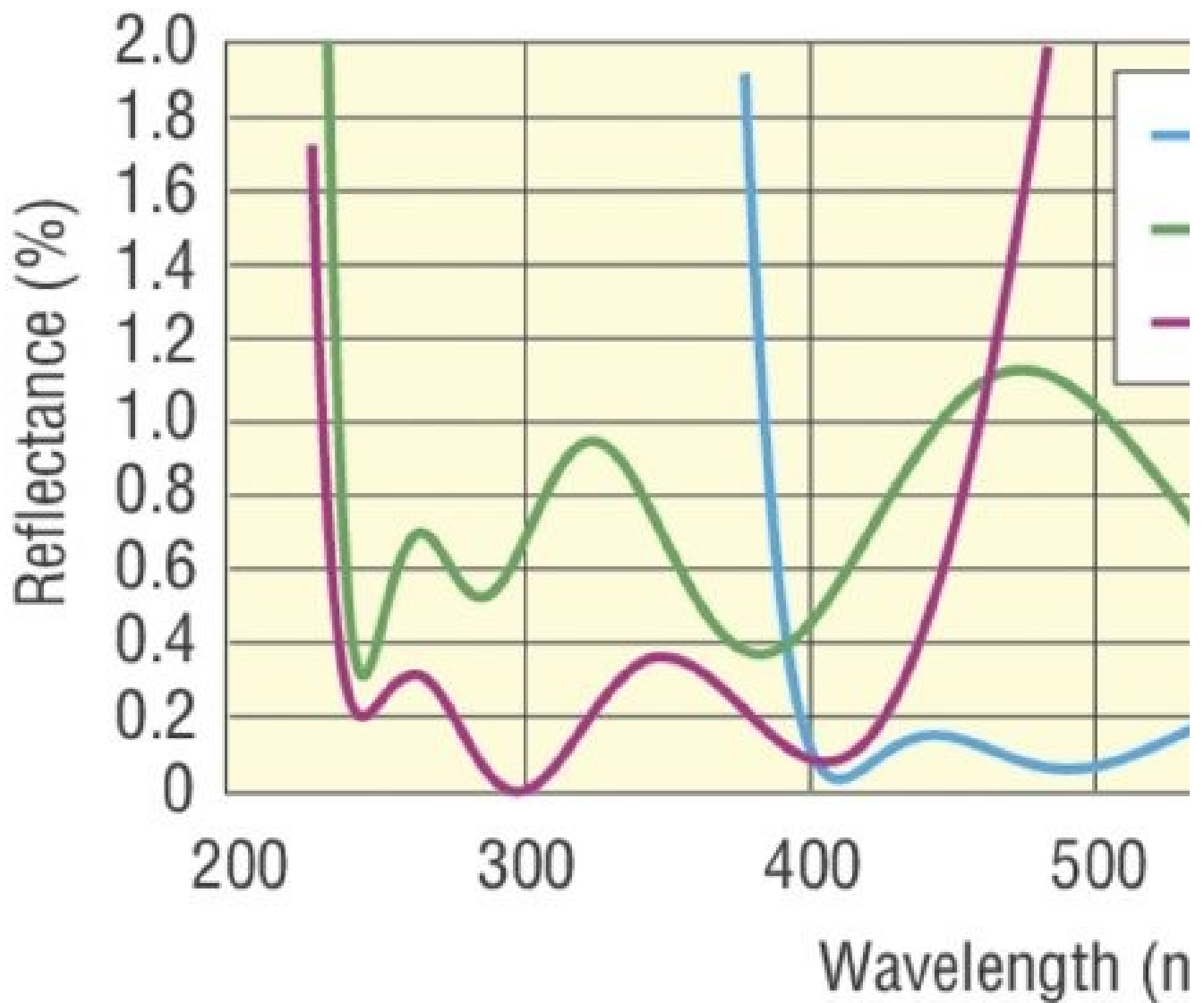


Figure 2. Typical broadband anti-reflection coating in the UV and VIS spectral regions.

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